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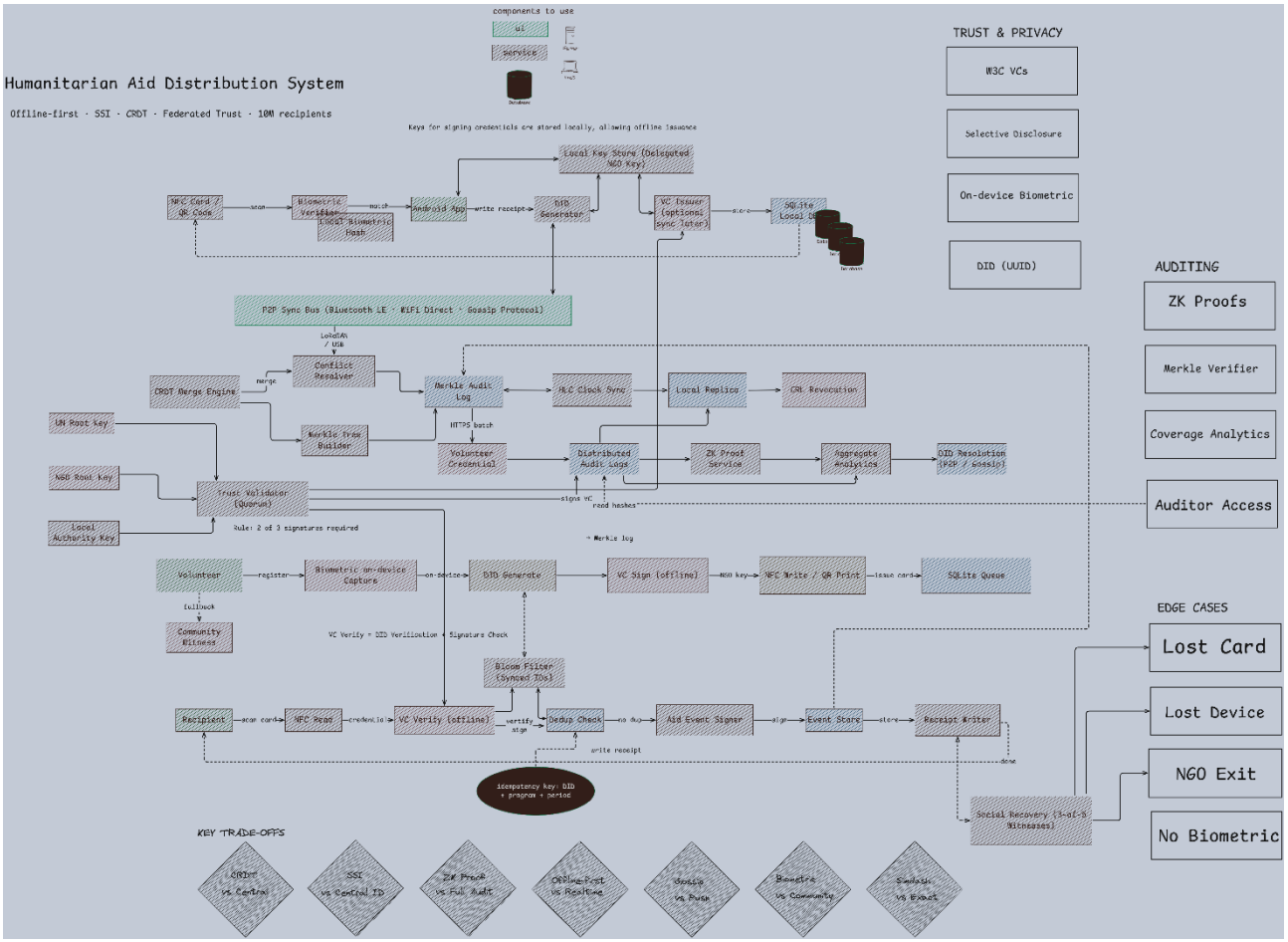

1) Problem Statement - Clarifying Questions:

During humanitarian crises, aid distribution suffers from duplication, fraud, and missed recipients - especially where identity documents are unavailable or unreliable. A coalition of NGOs and the United Nations require a digital system that enables fair, transparent aid delivery even in remote or fully offline regions.

Scale: Up to 10M recipients across all active crises simultaneously. Number of NGOs in coalition? 20-100 NGOs, each with 10-500 field volunteers. Connectivity in the field? Assume fully offline by default; sporadic 2G/3G available 10-30% of the time. Biometric modalities available? Fingerprint primary; iris scan secondary; facial recognition fallback; community witness as last resort. Data ownership model? Recipients own their own data; no NGO holds a central identity database. Audit access? Any accredited observer can audit aggregate stats; individual data stays private. Device availability? Android tablets with NFC reader; fallback to paper QR codes for recipients without devices.

2) High-Level Architecture:

The system is organized in three tiers, each designed to function independently when the tier above is unavailable.

Layer 1 - Field: Volunteer Android app, recipient NFC smart card / QR code, on-device biometric verifier. Connectivity: None - fully offline. Layer 2 - Regional hub: Raspberry Pi with solar power, CRDT merge engine, append-only Merkle audit log. Connectivity: Sporadic 2G/3G; syncs with field devices via BLE/WiFi Direct. Layer 3 - Global: Federated NGO trust anchors, UN root-of-trust, global audit log. Connectivity: Internet required; operates async from field operations.

3) Identity System: Self-Sovereign Identity:

The system implements the W3C Verifiable Credentials standard.

What is a Verifiable Credential? It is a cryptographically signed JSON document that proves a claim without requiring the issuer to be online.

1.Issuer: the NGO that registered the recipient.

2.Subject: the recipient.

3.Proof: a digital signature that any device can verify offline using the NGO’s public key.

Recipient Registration Flow:

1.Volunteer opens the Android app and selects Register new recipient.

2.Volunteer captures biometric template on-device.

3.App generates a DID for the recipient.

4.App issues a Verifiable Credential signed with the NGO’s private key.

5.Credential is written to the recipient’s NFC smart card or QR code.

6.Registration event is stored locally in SQLite and queued for sync.

Privacy Guarantees: Raw biometric data never leaves the device. Only the match result is used. Recipients can use selective disclosure to prove eligibility without revealing names or ages.

4) Offline-First Design - Synchronization:

The solution uses CRDTs to guarantee eventual consistency without a central authority.

1. Local Storage Each volunteer device runs a local SQLite database. Every write operation generates an event with a globally unique ID. The combination of recipient_did + program_id + period forms the idempotency key to prevent duplicate aid in the same cycle.

2. Synchronization Tiers Device <-> Device: Bluetooth Low Energy or WiFi Direct via Gossip protocol. Device -> Regional hub: USB cable or LoRaWAN. Hub acts as a relay. Regional hub -> Cloud: HTTPS batch upload when internet is available.

3. Clock Synchronization The system uses Hybrid Logical Clocks to guarantee causal ordering even when physical clocks are skewed between offline devices.

5) Trust Model: Federated

UN: Signs certificates for accredited NGOs. Has no access to recipient data. NGO: Issues Verifiable Credentials to recipients and signs aid events. Volunteer: Holds a credential signed by their NGO. Private key lives on the device. Recipient: Holds their own credential on their NFC card. Auditor: Can access the global Merkle audit log hashes.

6) Aid Delivery Flow:

1.Recipient presents NFC card or QR code.

2.App reads and verifies the Verifiable Credential signature offline.

3.App checks local database for duplicate delivery.

4.If no duplicate - volunteer confirms delivery and app creates a signed aid event.

5.App writes a delivery receipt back to the recipient’s card.

7) Fairness Auditing - Transparency:

1. Aggregate Analytics Coverage rate by region, duplicate detection, and gap analysis for missed recipients.

2. Zero-Knowledge Proofs The system uses ZK proofs to allow NGOs to prove enrollment counts or eligibility thresholds without revealing individual records or raw numbers.

3. Merkle Audit Log Every aid event is hashed and appended to a Merkle tree. Any auditor can verify that a specific event is included in the log by checking its Merkle proof without needing access to the full dataset.

8) Edge Cases - Failure Modes:

Biometric unavailable: Fallback to iris scan or two community witnesses. Volunteer device lost: Certificate revocation list updated on next sync. NGO leaves coalition: NGO certificate is revoked but historical records are preserved. Recipient loses card: Volunteer re-issues credential via biometric re-verification.

9) Scalability - Non-Functional Requirements:

10M recipients: DIDs are UUID-based with no central registry needed. Offline-first: All critical operations work with zero connectivity. Low-end devices: Runs on Android 8+ with 2GB RAM using TensorFlow Lite models.

10) Technology Choices:

Mobile app: Android Kotlin. Identity standard: W3C Verifiable Credentials and DIDs. CRDT library: Automerge. Logical clocks: Hybrid Logical Clocks. Regional hub: Raspberry Pi 4. ZK proofs: Bulletproofs.